Brady Bangasser

Site Reliability & Platform Engineer

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Systems engineer focused on reliability and platform work: running production HPC and cloud infrastructure, automating operations, and keeping large-scale systems available and observable under real load.

EXPERIENCE

System Administrator · Iowa State University — Apr 2025 – Present

- Operate and maintain a production HPC cluster with SLURM, Kubernetes, Active Directory, Ansible, Prometheus, and Apptainer, keeping it secure and available for a large user base.
- Built diagnostic and monitoring workflows that isolate hardware degradation, network bottlenecks, and software-stack faults before they take nodes offline.

Head of Networking · Cardinal Space Mining — Aug 2024 – Present

- Designed, built, and tested the internal network for a competition mining robot, ensuring redundant control and telemetry throughout each arena run.
- Collected multiple camera streams and relayed them to a third-party platform for live judging.

Cloud Security Engineer · John Deere — May 2025 – Aug 2025

- Built and deployed tooling that verified 100,000+ AWS and Azure resources against company policy, NIST standards, and platform best practices.
- Designed and managed least-privilege IAM policies in AWS to tighten cloud access control.

Research Assistant — High Performance Computing · Bethel University — Aug 2023 – Aug 2024

- Built and configured two bare-metal HPC clusters from scratch: networking, custom DNS and routing, user management, and SLURM job scheduling with VMware and Debian.
- Developed a self-healing IoT sensor mesh network to detect earthquakes, landslides, and flash floods in remote regions.

PROJECTS

Erid — Unified Cloud Images — 2026

- Single Packer + Terraform pipeline that builds identical, reproducible golden images across multi-cloud and on-prem targets, applying the same hardening everywhere to eliminate drift.
- Standardized a hardening baseline applied identically to every target image, so security posture no longer varies by platform.
- Replaced per-cloud manual image setup with one definition, cutting drift and setup time across environments.

Omni HTTP Router — 2025–2026

- Wrote a compiler that lowers a routing DSL directly to native object files, and compiles matching Terraform and container deployments alongside the app binary.
- Benchmarked the static router against conventional middleware stacks to quantify the latency and allocation savings.

Computer Vision Parking Enforcement Radar — 2025–2026

- Architected a Redis Streams pipeline for low-latency, high-throughput image and detection routing from a REST API to processing servers, backed by PostgreSQL geospatial storage.
- Tuned the broker for low-latency, high-throughput delivery from the REST API to processing servers.

EDUCATION

M.S. Computer Science · Iowa State University — Aug 2025 – May 2027

GPA: 4.00

- Concurrent B.S./M.S. program
- Coursework: Advanced Topics in High Performance Computing, Distributed Systems and Middleware, Data Security in Machine Learning, Network Design and Security

B.S. Computer Science · Iowa State University — Aug 2024 – Dec 2026

GPA: 3.79 · Dean's List

- Coursework: Compiler Design and Development, AI and Machine Learning in High Performance Computing, Network Architecture and Security, Theory of Computing, Software Design Practices

TECHNICAL SKILLS

Languages: C, C++, Python, Go, Rust, Bash, TypeScript
Infrastructure & Platform: Kubernetes, Docker, Terraform, Packer, Ansible, SLURM, Prometheus, Apptainer
Cloud: AWS, Azure, Google Cloud, AWS EKS
Data & Backends: PostgreSQL, Redis